

The empirical recurrence for OEIS A233630: recovering a conjecture that is not written down

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Abstract

OEIS A233630 records “Empirical recurrence of order 63”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 256$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 63, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 63 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233630 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with every 2×2 subblock having the sum of the absolute values of the edge differences equal to 6.”. It has offset 1 and begins

4974, 150054, 4350820, 130732438, 3912088926, 117571581104, 3530630987208, 106103463522318, ...

The entry states:

Empirical recurrence of order 63 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 08:53:32, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 63 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	27	c_{17}	7976063438633450	c_{33}	1043053492684463196859	c_{49}	1246189224950
c_2	615	c_{18}	−10776434181072390	c_{34}	−2142819932188386377834	c_{50}	−1300405897883
c_3	−14945	c_{19}	−88435030592140540	c_{35}	−1403732785275416979581	c_{51}	108864158818
c_4	−129196	c_{20}	133619548233547369	c_{36}	3017544044587744028462	c_{52}	1635330755246
c_5	3140161	c_{21}	741106921988150436	c_{37}	1436212245739424797069	c_{53}	−14674068294
c_6	12272509	c_{22}	−1207063909483124455	c_{38}	−3270918588011127764960	c_{54}	−137391406364
c_7	−340742407	c_{23}	−4742738018736919747	c_{39}	−1103704696284290194578	c_{55}	10456777947
c_8	−564565156	c_{24}	8151865794900815387	c_{40}	2712090114884712826359	c_{56}	73336732543
c_9	21994327203	c_{25}	23343023203870784677	c_{41}	626446948104008610941	c_{57}	2037438797
c_{10}	9199703317	c_{26}	−41817056523735788906	c_{42}	−1705764505381173549127	c_{58}	−2308029850
c_{11}	−919523599223	c_{27}	−88749181209873341378	c_{43}	−256294691718248544071	c_{59}	−473065392
c_{12}	301404418829	c_{28}	164585133629854003241	c_{44}	805200951156296043582	c_{60}	3739029333
c_{13}	26323431380254	c_{29}	261134122471621210207	c_{45}	72676504299837020737	c_{61}	155387054
c_{14}	−21226682019072	c_{30}	−500037378478303799180	c_{46}	−281500173982647788892	c_{62}	−22214813
c_{15}	−536024568630794	c_{31}	−59431269888884721638	c_{47}	−13233855588474962442	c_{63}	−12967654
c_{16}	604491515458256	c_{32}	1176200547921662555042	c_{48}	71668343454431730482		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 63, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $63 + 4$ terms removed returns order 63 as well, so $\deg q_{\min} = 63 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $63 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 63 that this sequence satisfies — and that family turns out to have exactly one member.

References

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